Tennessee Technological University

Department of Electrical and Computer Engineering

ECE 3050: Circuits II Laboratory

*Experiment 3 Report*

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Date of Experiment: 2/11/2026

Date of Submission: 3/4/2026

## *I. Introduction*

The purpose of Experiment 3 is to explore the transient responses of second-order electrical circuits. This lab will focus on the analysis of natural responses of series RLC circuits with varied resistances. The goal is to determine how different resistance values affect the damping behavior of a series RLC circuit by observing the transient response using an oscilloscope and comparing the results to theoretical calculations [3]. The varied resistors will allow examination of different damping behaviors that an RLC circuit can have such as critically damped, overdamped, and underdamped [3]. Upon completion, this lab will strengthen the understanding in the responses of second-order electrical circuits and how different resistances influence behavior.

## *II. Background/Theory/Design*

Unlike a first-order circuit, second-order circuits depend on two irreducible energy storage components. Irreducible refers to the idea that the circuit energy storage components have been reduced as much as possible without altering the circuit [1]. Second-order circuits are represented with a second-order differential equation in the following form:

$$\frac{d^2}{dt^2}x(t) + 2\alpha\frac{d}{dt}x(t) + \omega_0^2 x(t) = f(t)$$

where $x(t)$ is the circuit output function and $f(t)$ is the forcing function. In this experiment, the forcing function is 0, which denotes the natural response of an RLC circuit [1], [3]. Usually, the output function is defined by inductor current or capacitor voltage, depending on what is being solved for. The resonant frequency is defined by $\omega_0$, and the damping coefficient is denoted by $\alpha$ [3]. This lab report presents the analysis of a series RLC circuit. In the post-lab analysis and

discussion, the behaviors of series and parallel RLC circuits will be compared to understand the differences in the damping responses. For a series RLC circuit, the output is usually determined by capacitor voltage $v(t)$ [3]. The differential equation in this experiment becomes:

$$\frac{d^2}{dt^2}v(t) + \frac{R}{L}\frac{d}{dt}v(t) + \frac{1}{LC}v(t) = 0$$

The damping coefficient and resonant frequency values are critical for understanding what type of damping response will occur during the experiment. Another important variable is the critical damping resistance ($CDR$), which describes the resistance value that reduces overshoot in oscillation from a transient response back to a steady response as quickly as possible [1]. The damping coefficient, resonant frequency, and critical damping are defined as

$$2\alpha = \frac{R}{L} \quad \rightarrow \quad \alpha = \frac{R}{2L}$$

$$\omega_0^2 = \frac{1}{LC} \quad \rightarrow \quad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$R_{CDR} = 2\sqrt{\frac{L}{C}}$$

The damping coefficient and critical damping resistance equations will differ as these parameters are specific to a series RLC circuit design [1], [3]. The circuit is critically damped when $\alpha = \omega_0$, overdamped when $\alpha > \omega_0$, and underdamped when $\alpha < \omega_0$ [2], [3]. For more theory and information regarding second-order circuits, please refer to [1].

The design for this experiment will be a RLC series circuit using a function generator square wave as the voltage input. The square wave will have a DC offset of 2 $V$ and an amplitude of 2 $V$. A frequency of 25 $Hz$ will be fixed for all resistors with a duty cycle of 75 % [3]. The

input will be applied to all components in series with the resistor, capacitor, and inductor. Resistors will vary with values which will be 100 Ω, 470 Ω, 820 Ω, 2 $k\Omega$, and 10 $k\Omega$ [3]. A digital multimeter may be used to verify resistor tolerances during the experiment. The capacitor will be a value of $0.1\ \mu F$ and the inductor will be value of $22\ mH$ [3]. The voltage across the capacitor will be measured using an oscilloscope to analyze the transient response of the circuit when the square wave changes state. Figure 1 below shows the circuit design used in this lab in LTspice software. For more information regarding the equipment used, please refer to the Appendix.

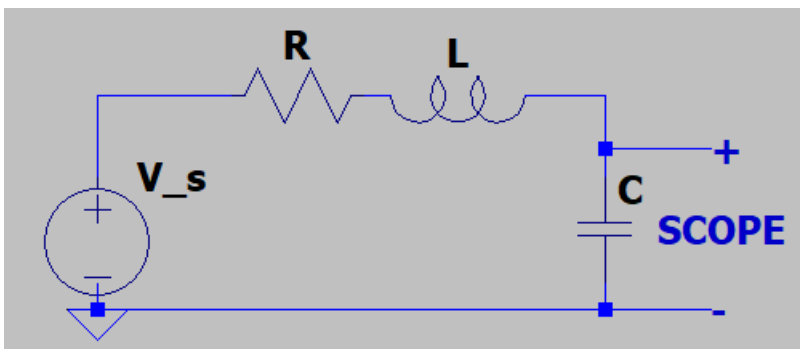


*Figure 1. RLC series circuit simulation with scope probe reference.*

Though it is not necessary for this experiment, it would be helpful before the experiment to compute the rise and fall times for each resistor used in the experiment. To do this, refer to the time constant $\tau$ [4]. Since the capacitor voltage rise and fall times will be measured, the following equation will be used:

$$\tau = RC$$

where R is the resistance and C is the capacitance. For this lab, the theory about $\tau$ is not critical, rather knowing the values to conduct measurements for each rise and fall time will be more important [4]. For the rise time of the capacitor, measure from $0\ V$ to roughly $2.528\ V$. For the

fall time of the capacitor, measure from $4\ V$ to approximately $1.472\ V$ [4]. It is vital to note that these values with be an approximation as this calculation is more specific to RC circuit design. The inductor in the circuit can induce errors in the rise and fall times that would not otherwise occur in a simple RC circuit design [3], [4].

## *III. Procedure*

### *Pre-lab Computations*

Before recording any lab measurements in the lab, use LTspice to simulate Figure 1 for all resistors in this lab. Save these simulation results to verify lab results. Build a table that will record the resistor value, damping coefficient $\alpha$, resonant frequency $\omega_0$, critical damping resistance ($CDR$), determination of whether the value of $R$ is greater than the $CDR$, damping type, and rise and fall times of the capacitor voltage [3]. Compute the resonant frequency, critical damping resistance, and damping coefficient for each resistor used in the lab. Note: the resonant frequency will be the same for all resistors [3]. Determine the damping types for each resistor in the circuit. Also, as stated before, it is recommended to compute the rise and fall times of the capacitor voltage prior to measuring the rise and fall times in the experiment to assure each value is roughly equivalent to the calculations [4]. These values will not be the same, but it is a good reference during the experiment.

### *Lab Preparation*

Once ready to perform the experiment, ensure all instruments are working accordingly and calibrated properly. Do not activate any equipment before all connections are made and verified. If needed, use a digital multimeter to confirm resistance values. Use a solderless breadboard to assemble the resistor, inductor, and capacitor in series. Check for a good

connection with each component and attach the positive terminal of the function generator to the resistor and the negative terminal to the end of the capacitor. When using an electrolytic capacitor, verify the polarity before energizing any equipment. Attach the positive lead of the oscilloscope to the positive terminal of the capacitor as well as the negative lead to the common ground. Verify the oscilloscope probe is switched to the proper attenuation setting of the oscilloscope. If applicable, refer to the oscilloscope probe's data sheet for more information. Configure the function generator with the parameters mentioned in the design. Attach trackable cursors to the oscilloscope display to begin measurements. If clarification is needed regarding component connections, please refer to Figure 2 below.

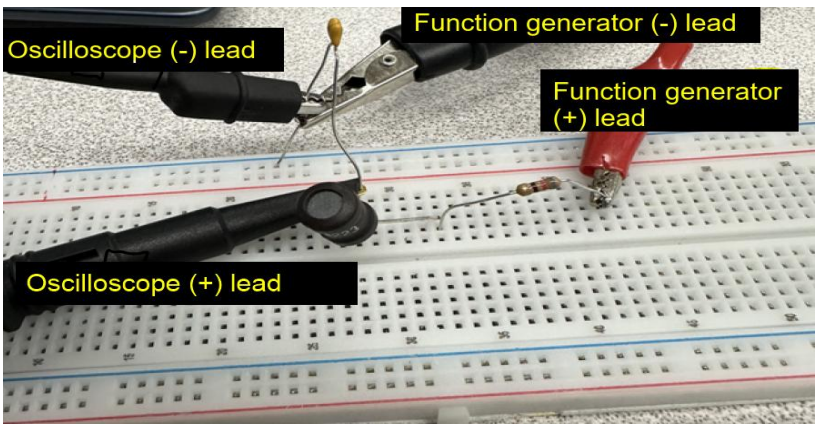


*Figure 2. RLC series circuit on solderless breadboard.*

***Measurements***

Turn on the function generator output and ensure the transient response is visible on the oscilloscope. If not, use the oscilloscope knobs to adjust the vertical scale until the waveform is properly displayed. Use the oscilloscope cursors to measure both rise and fall times of the capacitor voltage during transient responses. Verify accurate measurements and record these values in the lab data sheet. Observe the waveform for each resistor and check the damping type

(underdamped, critically damped, or overdamped). Confirm the experimental results correspond with the pre-lab data. Record pictures for the rise and fall times of each circuit. Once each rise and fall time is recorded, scale out the oscilloscope and take a picture of the entire waveform [3].

***Disassembly***

After completing the experiment, turn off all equipment and disconnect the circuit. Clean the workspace and store all equipment and leads properly before leaving the lab.

## *IV. Results, Analysis, Discussion*

**Table 1: RC Response Results**

| *Resistor (Ω)* | *Value of $\alpha$ (rad*s$^{-1}$)* | *Value of $\omega_0$ (rad*s$^{-1}$)* | *Is R greater than CDR?* | *Damping Type* | *Rise Time ($\mu s$)* | *Fall Time ($\mu s$)* |
|---|---|---|---|---|---|---|
| 100 | 2,273.0 | 21,320.0 | No | Underdamped | 44.0 $\mu s$ | 61.0 $\mu s$ |
| 470 | 10,682.0 | 21,320.0 | No | Underdamped | 57.0 $\mu s$ | 66.0 $\mu s$ |
| 820 | 18,636.0 | 21,320.0 | No | Underdamped | 82.0 $\mu s$ | 78.0 $\mu s$ |
| 2,000 | 45,455.0 | 21,320.0 | Yes | Overdamped | 204.0 $\mu s$ | 180.0 $\mu s$ |
| 10,000 | 227,273.0 | 21,320.0 | Yes | Overdamped | 1,200.0 $\mu s$ | 1,080.0 $\mu s$ |

The table above describes the recorded data of the experiment as the procedure explained. The results of this experiment were as intended. The LTspice simulation used for the first resistor

of 100 Ω matched the oscilloscope output waveform of an underdamped response. Below is an attached waveform of the oscilloscope (in Figure 3).

*Figure 3. Waveform of the series RLC when R = 100 Ω.*

In this circuit, the computed resonance frequency $\omega_0$, 21,320.0 rad*s$^{-1}$, was greater than the damping coefficient $\alpha$, which was calculated to be 2,273.0 rad*s$^{-1}$. Based on the theory of RLC circuit design, when $\omega_0^2$ (resonant frequency) is larger than $\alpha^2$ (damping coefficient), the circuit has an underdamped response [2]. An underdamped circuit generates greater oscillation for each steady state change of the circuit. The oscillation of the waveform occurs multiple times until steady state voltage is reached [1]. When the damping coefficient $\alpha$ is small, the resistor $R$ is small as denoted by the equation

$$\alpha = \frac{R}{2L}$$

where the circuit is a series RLC design [1]. Due to the lower resistance, the oscillations are more noticeable and larger than if a higher resistance was used [2]. This will be discussed later with the other resistors. Furthermore, the biggest discrepancy of this circuit was during the measurements of the rise and fall times of the capacitor voltage. The computed circuit rise and

fall time of 1 time constant were 10.0 $\mu s$. However, the lab measurements for the rise and fall times were 44.0 $\mu s$ and 61.0 $\mu s$, respectively. The LTspice simulation shared similar results to measured rise and fall times for 1 time constant. In Figures 4 and 5 below, the oscilloscope pictures share the capacitor voltage rise and fall times of 1 $\tau$ when $R$ is equal to 100 Ω.

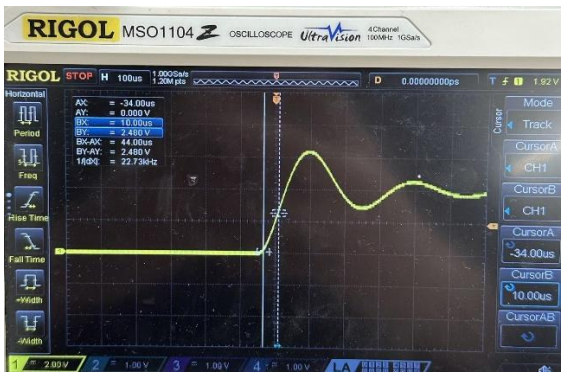


*Figure 4. Rise time of the series RLC when R = 100 Ω.*

*Figure 5. Fall time of the series RLC when R = 100 Ω.*

While the theoretical rise and fall time at 1 time constant was calculated at 10.0 $\mu s$, the measured laboratory values were significantly higher. This discrepancy is likely contributed to the non-ideal characteristics of the equipment. The internal resistance of 50 Ω from the function generator and the inductor resistor could have increased the damping time of the system. The resistance being so low with only a value of 100 Ω could have made these characteristics more apparent, therefore, extending the time for the capacitor to reach 1 time constant.

For the other resistors used in this experiment, the measured rise and fall times were comparable to the calculated rise and fall times. For the next resistance used, the value was 470 Ω. The calculated rise and fall times of the capacitor voltage at 1 time constant are both 47.0 $\mu s$. The measured rise and fall times were 57.0 $\mu s$ and 66.0 $\mu s$, respectively. Again, this error could have resulted due to the non-ideal characteristics of the lab equipment. Inductor's series resistance as well as the function generator internal resistance could have increased these

measurements slightly. The inductor could also alter the timing as it is not involved in rise and fall time calculations. The rise and fall times that are being measured are more specific to an RC circuit and therefore could present such error [4]. Furthermore, these values are very close for a non-ideal setting. Figures 7, 8, and 9 below share the oscilloscope results.

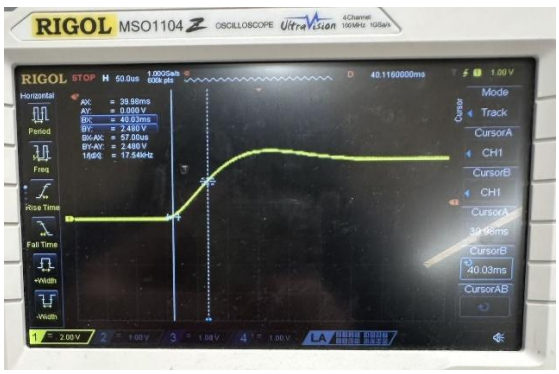


*Figure 7. Rise time of the series RLC when R = 470 Ω.*

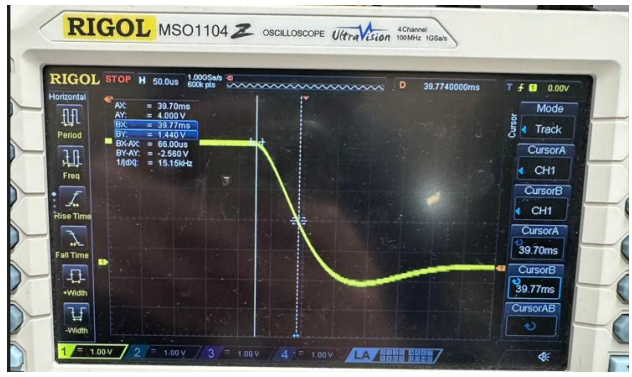


*Figure 8. Fall time of the series RLC when R = 470 Ω.*

*Figure 8. Waveform of the series RLC when R = 470 Ω.*

Based off pre-lab calculations, the damping coefficient $\alpha$ is 10,682.0 rad*s$^{-1}$, and the resonant frequency $\omega_0$ is the same as before with a value of 21,320.0 rad*s$^{-1}$. Theory states when ${\omega_0}^2$ is larger than $\alpha^2$, the circuit will be underdamped as stated before [1], [2]. However, it is important to recognize with larger resistance, it alters the transient response. As seen in Figure 8, the oscillations are less apparent in comparison to when $R$ was equal to 100 Ω. As the resistance approaches the critical damping resistance ($CDR$), the circuit quickly reaches steady state in the

quickest time without any oscillation [1]. The next resistor used in this lab experiment was an 820 Ω resistor. Figures 9, 10, and 11 below display the results of the rise and fall times as well as a general photo of the waveform.

*Figure 9. Rise time of the series RLC when R = 820 Ω.*

*Figure 10. Fall time of the series RLC when R = 820 Ω.*

*Figure 11. Waveform of the series RLC when R = 820 Ω.*

The circuit with the 820 Ω resistor has a calculated rise and fall time of 82.0 $\mu s$. The measured rise and fall times were 82.0 $\mu s$ and 78.0 $\mu s$, respectively. These measured values are approximate to the computed values. The 820 Ω resistor shares the most important information regarding the damping response in this entire circuit. This resistor is the closest to the critical damping resistance, however, it is still underdamped slightly with a damping coefficient of 18,636.0 rad*s$^{-1}$ and a resonant frequency of 21,320.0 rad*s$^{-1}$ [1], [2]. The critical damping resistance is 938 Ω while the resistor is right below this threshold. This allows examination of a

waveform that closely resembles a critically damped circuit response (when $\omega_0^2 = \alpha^2$) [1], [2]. The waveform in Figure 11 roughly describes a waveform with no apparent oscillation when the circuit rises and falls for each steady state. This is important because it illustrates the ideal equilibrium of a transient response in a RLC circuit [1], [3]. The next two resistors will determine another damping response that has yet to be discussed.

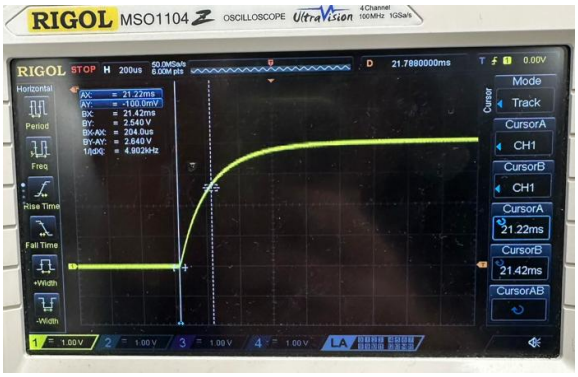


*Figure 12. Rise time of the series RLC when R = 2 kΩ.*

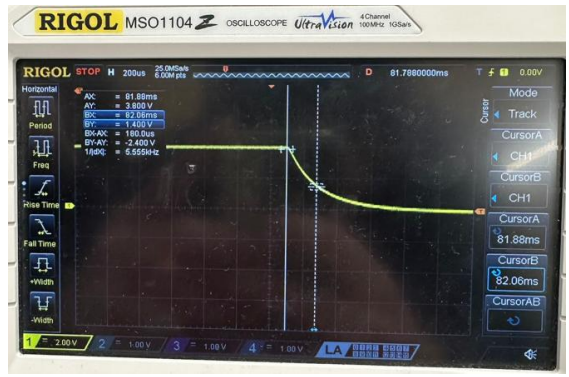


*Figure 13. Fall time of the series RLC when R = 2 kΩ.*

*Figure 14. Waveform of the series RLC when R = 2 kΩ.*

In Figure 12 and 13, the rise and fall times of the series RLC circuit are shared when $R = 2\ k\Omega$. The computed rise and fall time for 1 time constant is $200.0\ \mu s$. The measured rise and fall times were $204.0\ \mu s$ and $180.0\ \mu s$, respectively. The rise and fall times are appropriate in comparison to the calculated rise and fall time due to the equipment having non-ideal characteristics. However, there is a slight difference in the pattern of the resonant frequency and

damping coefficient. The damping coefficient $\alpha$ for this RLC series circuit is 45,455.0 rad*s$^{-1}$, while the resonant frequency $\omega_0$ remains the same value of 21,320.0 rad*s$^{-1}$. Based on second-order circuit theory, $\omega_0^2$ is larger than $\alpha^2$, therefore, the circuit will now be overdamped versus underdamped [1]–[3]. Overdamped circuits refer to no visible oscillation, but a noticeable time interval when the circuit changes from steady state [1]. Refer to Figure 14 above for the overdamped response. This response closely resembles a first-order RC circuit, where the transient response is noticeable in each steady state transition. In Figure 15, 16, and 17, the final resistor is used with a value of 10 $k\Omega$. The overdamped response of the 10 $k\Omega$ resistor is more visible than the 2 $k\Omega$ resistor.

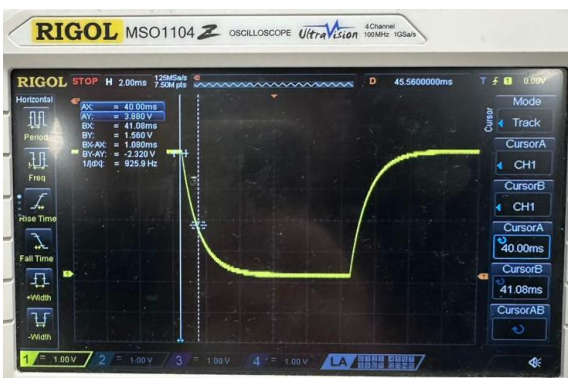


*Figure 15. Rise time of the series RLC when R = 10 kΩ.*

*Figure 16. Fall time of the series RLC when R = 10 kΩ.*

*Figure 17. Waveform of the series RLC when R = 10 kΩ.*

The rise and fall times of the series RLC circuit with the resistor equal to $10\ k\Omega$ was computed to be $1{,}000.0\ \mu s$. The measurements taken in the experiment for the rise and fall times of $1\ \tau$ were $1{,}080.0\ \mu s$ and $1{,}200.0\ \mu s$, respectively. Again, the rise and fall times are consistent with the computed values, with minor discrepancies due to the equipment having non-ideal characteristics. The overall waveform clearly shows a greater time per new steady state condition for each rise and fall of the capacitor voltage. Consequently, it can be inferred that the circuit's time constant with each resistance are directly proportional.

Overall, the observed computations are consistent with the measurement results. As the resistance is increased, the oscillation behavior of the RLC circuit decreases. The circuit transitions from a heavily underdamped circuit with a $100\ \Omega$ resistor to a near critically damped circuit with a $820\ \Omega$ resistor. A moderately underdamped circuit is established with the $470\ \Omega$ resistor, allowing for the observation of differences in each circuit. Furthermore, the current may be measured instead of voltage on the oscilloscope by placing a very small resistor in series after the component. Due to Ohm's Law, the voltage waveform will appear as an equivalent current waveform. However, an issue with this method is that extra resistance could introduce error in the measurements of other components from the circuit. Another issue could be the resistor's power rating. This method would be limited by the power rating of the resistor, and excessive current could introduce overheating.

The physical analysis of this experiment involved a series RLC circuit. Critically damped, overdamped, and underdamped parallel RLC circuit responses will be analyzed via LTspice software [3]. Figure 18 below shows the design of the parallel RLC circuit.

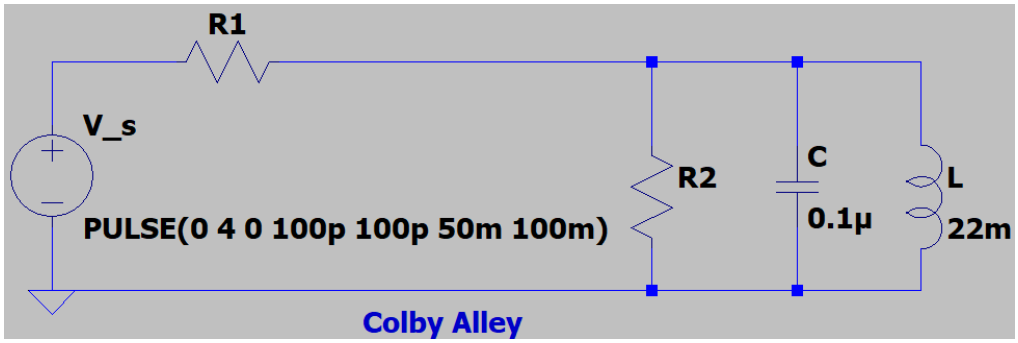


*Figure 18. RLC parallel circuit simulation.*

First, it is important to find the critical resistance to determine where critically damping will occur. Based on the circuit above, the critical damping resistance will be the equivalent resistance of $R_1$ and $R_2$. Since the voltage source is $0\ V$, $R_1$ and $R_2$ will be in parallel. According to the equations for parallel RLC circuits, the critical damping resistance is expressed by

$$R_{CDR} = \frac{1}{2}\sqrt{\frac{L}{C}}$$

where $L = 22\ mH$ and $C = 0.1\ \mu F$ [3]. Solving the critical damping resistances gives a value of approximately 235 Ω. For simplicity, denote $R_1 = R_2$. Conveniently, the lab kit offers a resistor value of 470 Ω, which when in parallel with an equivalent resistor gives $R_{eq} = 235\ \Omega$. Therefore, use two 470 Ω resistors for $R_1$ and $R_2$. Figure 19 below shows the waveform of a parallel RLC circuit that is critically damped.

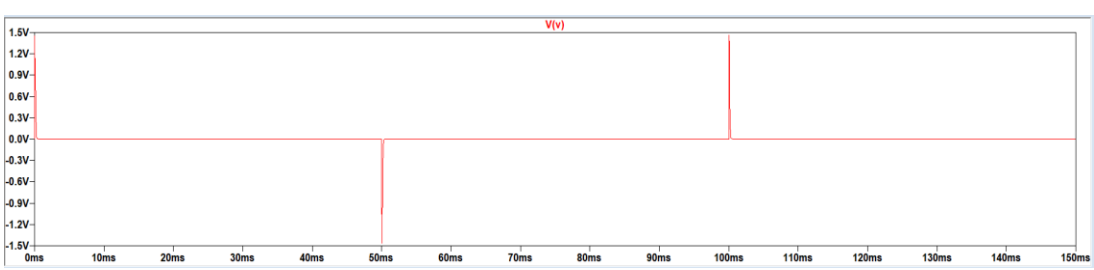


*Figure 18. RLC parallel circuit with a critically damped response.*

There are not any noticeable oscillations in this circuit, confirming a critically damped circuit response. Parallel RLC circuits are inversely related to series RLC circuits with respect to the critical damping resistance [1], [2]. A smaller resistance will be an overdamped parallel RLC circuit while a larger resistance will be an underdamped parallel RLC circuit, in contrast to a series RLC circuit [2]. Using two 150 Ω resistors for $R_1$ and $R_2$ will be used to achieve an equivalent resistance of 75 Ω. Figure 19 shows the same parallel RLC circuit with these values to demonstrate an overdamped circuit.

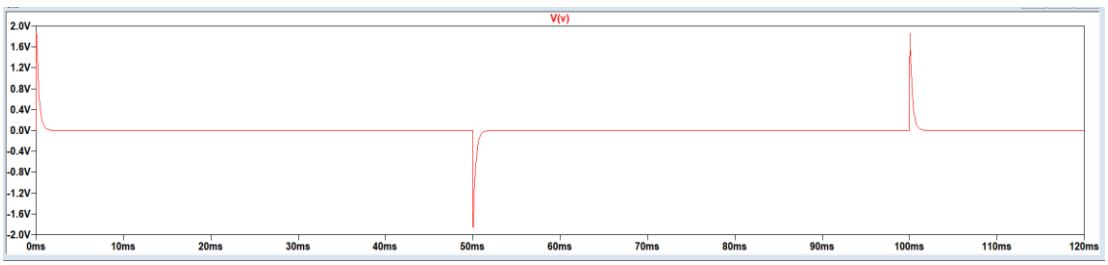


*Figure 19. RLC parallel circuit with an overdamped response.*

Figure 19 concludes that there is a slower response for each new steady state [1], [2]. Now, using two parallel 8.2 $k\Omega$ for values of $R_1$ and $R_2$ will give an underdamped response. Figure 20 below illustrates the waveform of this circuit.

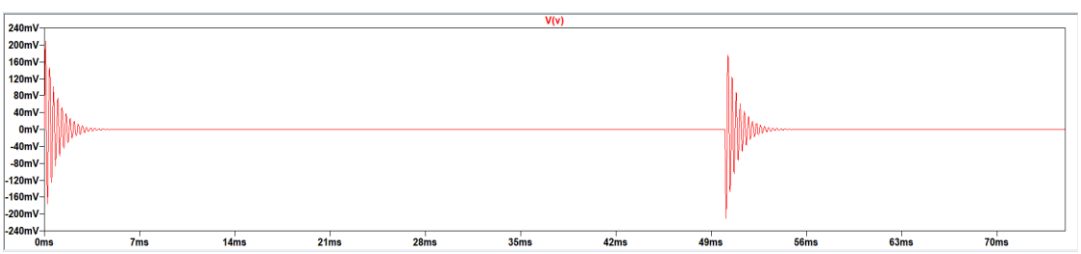


*Figure 20. RLC parallel circuit with an underdamped response.*

This waveform clearly describes an underdamped response with multiple noticeable oscillations before reaching the next steady state [1]. The comparison of a series and parallel RLC circuit allows for a better understanding of how each circuit behaves and how they are inversely related to resistances.

## *V. Summary and Conclusion*

The lab objective was successfully met through the investigation of an RLC series circuit with varying resistances. Using the procedure described above, this experiment demonstrated underdamped and overdamped conditions with resistors in the range from $100\ \Omega$ to $10\ k\Omega$. Notably, a circuit at $820\ \Omega$ displayed a near critically damped response with minor delay and no apparent oscillations. The observation before the lab experiment corresponded with the measured results during the experiment with minor discrepancies due to non-ideal equipment characteristics and component tolerances, which were discussed in the analysis above. The damping responses for each resistance were confirmed and validated by the background theory of second-order electrical circuits [1], [3].

## *VI. References*


[1] J. A. Svoboda, *Introduction to Electric Circuits*, 9th ed. Hoboken, NJ, USA: Wiley, 2018.

[2] C. Johnson. ECE 3050. Class Lecture, Topic: "Second Order Natural Circuit Response." Department of Electrical and Computer Engineering, Tennessee Technological University, Cookeville, TN, Jan. 20, 2026.

[3] Circuit and Electronics Lab, *Lab Manual 3: Damping of RLC Circuits*, Tennessee Technological University, 2026.

[4] Circuit and Electronics Lab, *Lab Manual 2: Time Constant of RC and RL Circuits*, Tennessee Technological University, 2026.

## *VII. Appendix*

**Table 2: Equipment**

| *Manufacturer* | *Model* | *Serial Number* | *Device* |
|---|---|---|---|
| Siglent | SDG2042X | SDG2XCAX4R0509 | Function Generator |
| Keysight | 34461A | MY59002200 | Digital Multimeter |
| Rigol | MSO1104 | DS1ZC171600509 | Oscilloscope |
| Agilent | HP9100 | – | Oscilloscope Probe |
| – | Generic | – | Waveform Output Cables |
| – | Generic | – | Solderless Breadboard |
| – | Generic | – | Resistors<br>(100 Ω, 470 Ω, 820 Ω, 2 $k\Omega$,<br>10 $k\Omega$) |
| – | Generic | – | Inductor (22 mH) |
| – | Generic | – | Capacitor (0.1 μF) |

***Prelab*** **Colby Alley Section 101**

**Lab Partner: Alek Presley**

We are familiar with the derivations of values for the RLC series circuit. Using the following equations:

Damping coefficient equation: $\alpha = \frac{R}{2L}$

Resonant frequency equation: $\omega_0^2 = \frac{1}{LC}$

Critical damping resistance equation for series RLC circuit: $R = 2\sqrt{\frac{L}{C}}$

Damping type for the 100 Ω resistor:

$$\alpha = \frac{R}{2 * L} \quad \rightarrow \quad \frac{100}{2 * (22 * 10^{-3})} \quad \rightarrow \quad \alpha = 2{,}273.0\, rad * s^{-1}$$

$$\omega_0^2 = \frac{1}{LC} \quad \rightarrow \quad \omega_0^2 = \frac{1}{(22 * 10^{-3})(0.1^{-6})} \quad \rightarrow \quad \omega_0 = 21{,}320.0\, rad * s^{-1}$$

$$CDR = 2 * \sqrt{\frac{L}{C}} \quad \rightarrow \quad CDR = 938.0 > R\ (100\ \Omega)$$

Since $\omega_0 > \alpha$, the circuit will be underdamped.

The critical damping resistance is larger than the resistance. Since the resonant frequency is dependent on the inductor and capacitor, the value will not change from 21,320.0 rad*$s^{-1}$. This is also true for the critical damping resistance value.

Damping type for the 470 Ω resistor:

$$\alpha = \frac{R}{2 * L} \quad \rightarrow \quad \frac{470}{2 * (22 * 10^{-3})} \quad \rightarrow \quad \alpha = 10{,}682.0\, rad * s^{-1}$$

$$\omega_0 = 21{,}320.0\, rad * s^{-1}$$

CDR = 938.0 > R (470 Ω)

Again, $\omega_0 > \alpha$, so the circuit will be underdamped. Resistor value is greater than the critical damping resistance.

Damping type for the 820 Ω resistor:

$$\alpha = \frac{R}{2 * L} \quad \rightarrow \quad \frac{820}{2 * (22 * 10^{-3})} \quad \rightarrow \quad \alpha = 18{,}636.0\, rad * s^{-1}$$

$$\omega_0 = 21{,}320.0\, rad * s^{-1}$$

CDR = 938.0 > R (820 Ω)

$\omega_0 > \propto$, the circuit with the 820 Ω resistor will be underdamped. The critical damping resistance remains larger than the resistor value again.

Damping type for the 2 kΩ resistor:

$$\alpha = \frac{R}{2 * L} \quad \rightarrow \quad \frac{(2 * 10^{3})}{2 * (22 * 10^{-3})} \quad \rightarrow \quad \alpha = 45{,}455.0\, rad * s^{-1}$$

$$\omega_0 = 21{,}320.0\, rad * s^{-1}$$

CDR = 938.0 < R (2 kΩ)

This time we have $\omega_0 < \propto$, therefore the circuit will be overdamped. The critical damping resistance is now smaller than the resistor value.

Damping type for the 10 kΩ resistor:

$$\alpha = \frac{R}{2 * L} \quad \rightarrow \quad \frac{(10 * 10^3)}{2 * (22 * 10^{-3})} \quad \rightarrow \quad \alpha = 227{,}273.0\, rad * s^{-1}$$

$$\omega_0 = 21{,}320.0\, rad * s^{-1}$$

CDR = 938.0 < R (10 kΩ)

Once again, $\omega_0 < \alpha$, so the circuit with this configuration will be overdamped. The critical damping resistance will be smaller than the resistor value.

For the LTspice simulation, the period and pulse width will be calculated using the following equations:

$$T = \frac{1}{f}$$

$$PW = DT$$

where $D$ is the duty cycle, $T$ is the period, $PW$ is the pulse width, and $f$ is the frequency used in the circuit. The frequency is given to be 25 $Hz$. The voltage source will be a square wave with an amplitude of 2 $V$ and DC offset of 2 $V$. The duty cycle will be set at 75 % as provided in the lab handout.

$$T = \frac{1}{25} \quad \rightarrow \quad 0.04 \quad \rightarrow \quad T = 40.0 \;\; ms$$

$$PW = DT \quad \rightarrow \quad \frac{75}{100} * (40 * 10^{-3}) \quad \rightarrow \quad PW = 30.0\, ms$$

It is helpful to compute rising and falling times for 1 time constant for the experiment. The rise time for 1 time constant is from 0 $V$ to approximately 2.53 $V$. The fall time for 1 time

constant is from $4\ V$ to approximately $1.472\ V$. Note: the value computed below might be altered during measurements due to this being a series RLC circuit and not just an RC circuit. Using the equation and values below we can compute for each rise and fall time expected.

$$\tau = RC$$

Value of $\tau$ for the $100\ \Omega$ resistor:

$$\tau \quad \rightarrow \quad 100 * (0.1 * 10^{-6}) \quad \rightarrow \quad \tau = 10.0\ \mu s$$

Value of $\tau$ for the $470\ \Omega$ resistor:

$$\tau \quad \rightarrow \quad 470 * (0.1 * 10^{-6}) \quad \rightarrow \quad \tau = 47.0\ \mu s$$

Value of $\tau$ for the $820\ \Omega$ resistor:

$$\tau \quad \rightarrow \quad 820 * (0.1 * 10^{-6}) \quad \rightarrow \quad \tau = 82.0\ \mu s$$

Value of $\tau$ for the $2k\ \Omega$ resistor:

$$\tau \quad \rightarrow \quad (2 * 10^{3}) * (0.1 * 10^{-6}) \quad \rightarrow \quad \tau = 200.0\ \mu s$$

Value of $\tau$ for the $10k\ \Omega$ resistor:

$$\tau \quad \rightarrow \quad (10 * 10^{3}) * (0.1 * 10^{-6}) \quad \rightarrow \quad \tau = 1{,}000.0\ \mu s$$

Attached below are the figures for the LTspice simulations and waveforms for each circuit as instructed in the handout.

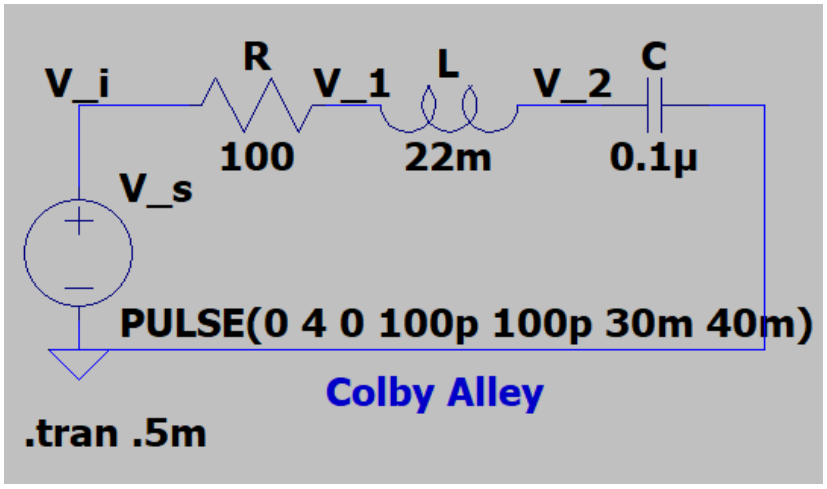


*Figure 1. LTspice RLC series circuit with 100 Ω.*

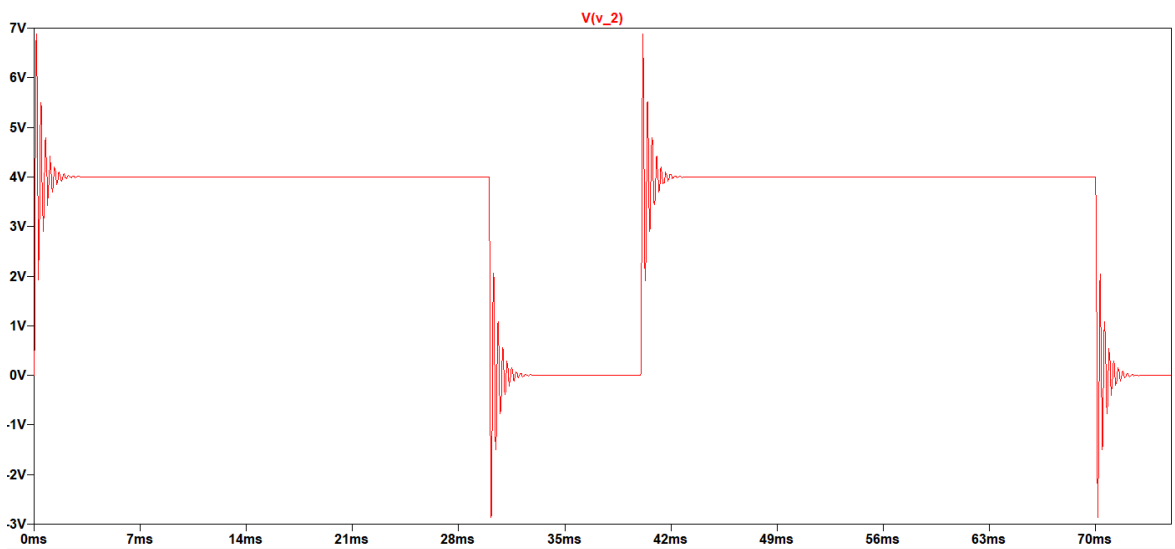


*Figure 2. LTspice waveform for RLC series circuit at $V_C$ with 100 Ω (underdamped).*

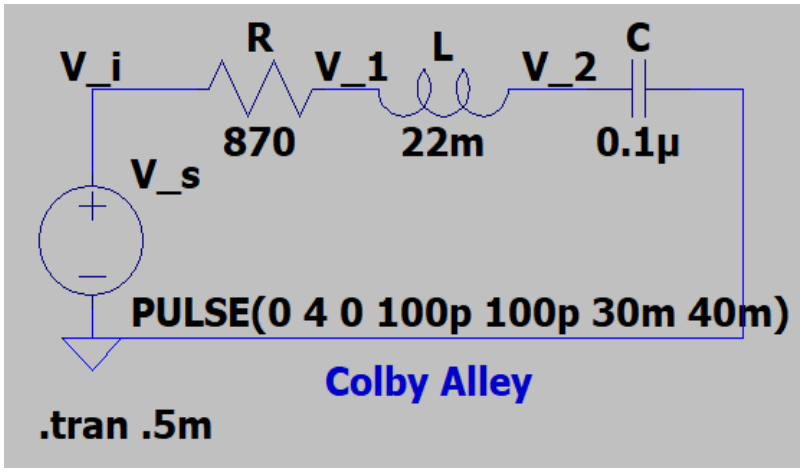


*Figure 3. LTspice RLC series circuit with 870 Ω.*

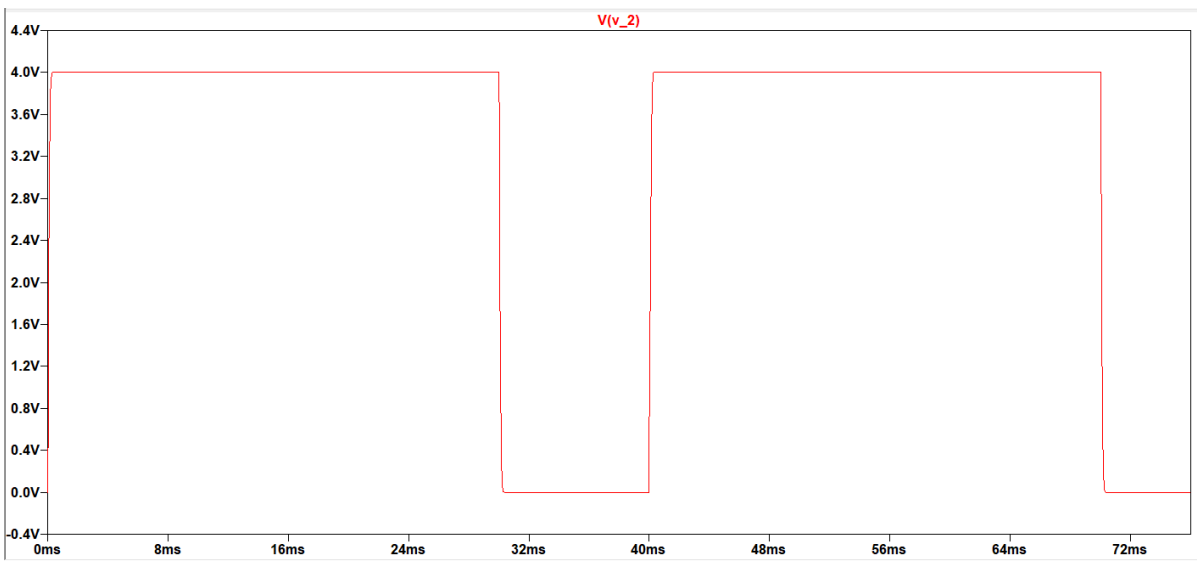


*Figure 4. LTspice waveform for RLC series circuit at $V_C$ with 870 Ω (underdamped).*

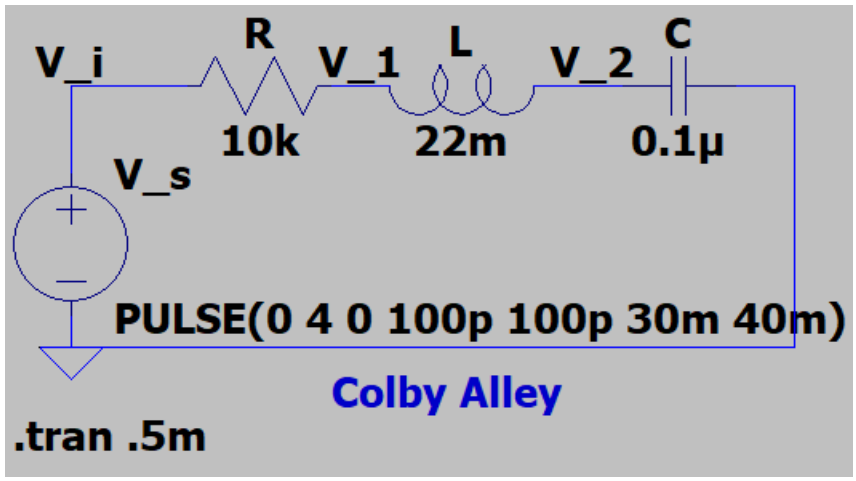


*Figure 5. LTspice RLC series circuit with 10 kΩ.*

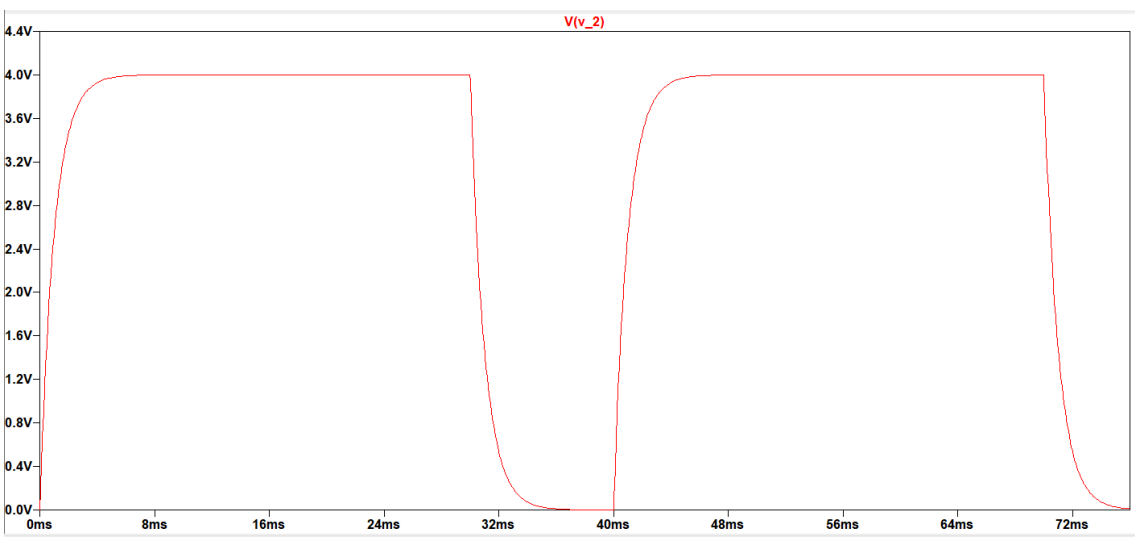


*Figure 6. LTspice waveform for RLC series circuit at $V_C$ with 10k Ω (overdamped).*

Attached below is the completed Table 1 for the RC response results based off the lab handout. This table includes the rise and fall times for the capacitor voltage.

**Table 1: RC Response Results**

| *Resistor (Ω)* | *Value of $\alpha$ (rad*s$^{-1}$)* | *Value of $\omega_0$ (rad*s$^{-1}$)* | *Is R greater than CDR?* | *Damping Type* | *Rise Time ($\mu s$)* | *Fall Time ($\mu s$)* |
|---|---|---|---|---|---|---|
| 100 | 2,273.0 | 21,320.0 | No | Underdamped | 44.0 $\mu s$ | 61.0 $\mu s$ |
| 470 | 10,682.0 | 21,320.0 | No | Underdamped | 57.0 $\mu s$ | 66.0 $\mu s$ |
| 820 | 18,636.0 | 21,320.0 | No | Underdamped | 82.0 $\mu s$ | 78.0 $\mu s$ |
| 2,000 | 45,455.0 | 21,320.0 | Yes | Overdamped | 204.0 $\mu s$ | 180.0 $\mu s$ |
| 10,000 | 227,273.0 | 21,320.0 | Yes | Overdamped | 1,200 $\mu s$ | 1,080.0 $\mu s$ |

Signature: _____

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